QUACKWORKS ACOUSTIC LOCK (QAL)

NON-CONFIDENTIAL OVERVIEW

Zero-BOM Software-Defined Second Line of Defence Against Keyless Vehicle Theft

UK IPO Patent Application No. GB2619997.6 (Claims 1-25) | Priority Date Established | Quackworks Audio IP

The Threat Landscape: Breaking the Radio & Bus Attack Vectors

Modern keyless vehicles can be stolen in under sixty seconds using RF radio relay kits or CAN-injection tools tapped into exterior wiring (such as headlight connectors). Existing countermeasures (RF fobs, BLE digital keys, and Ultra-Wideband radar) share the same vulnerability: they operate on external radio frequencies that can be amplified, bridged, or manipulated. Furthermore, passive proximity systems offer zero protection when physical keys and smartphones are stolen together from gym lockers, hotel rooms, or domestic break-ins.

The Architecture: In-Cabin Acoustic Air-Gap

QAL introduces an additive, physical-presence authentication factor that operates alongside a vehicle's existing BCM and keyless systems without requiring replacement. Once the driver is seated, an authenticated, dynamic near-ultrasonic acoustic token emitted by their smartphone is verified by the vehicle's pre-existing hands-free microphones before powertrain contactors or starter relays are permitted to close. Because acoustic pressure waves are physically confined by laminated automotive glass and door weather-stripping, the token cannot be intercepted, boosted, or relayed from outside the vehicle.

Key Commercial & Engineering Advantages

- **Zero Added Vehicle Hardware (£0 / $0 BOM):** Reuses pre-existing cabin microphones and audio DSP already installed across vehicle lines.
- **Immune to RF Relays & CAN Injection:** Sound waves cannot be bridged by radio amplifiers or spoofed via physical CAN-bus taps.
- **Biometric Proof-of-Intent:** Smartphone initiation (FaceID/PIN) prevents drive-aways even if keys and phone are stolen together.
- **Single-Use Anti-Replay Tokens:** Ephemeral nonces and temporal epoch validity prevent eavesdropping or audio playback attacks.
- **100% User-Optional "Driveway Shield":** Operates as an opt-in toggle in the infotainment menu for overnight driveway immunity.
- **Multi-Role Access Lifecycle:** Built-in support for revocable guest, valet, workshop maintenance, and dealership transit modes.
- **Forensic Audit Ledger:** Cryptographically hash-chained (SHA-256) logging aligned with ISO/SAE 21434 and UN-ECE WP.29 R155.
- **Functional Safety (ISO 26262):** Enforces an ASIL stand-still interlock (v <= 0.1 km/h), entering fail-silent dormancy while rolling.

Current Technology Maturity & Empirical Validation

The complete end-to-end architecture (including continuous-phase modulation, cryptographic token state logic, and acoustic demodulation) has been built and verified in a working Software-in-the-Loop (SIL) simulation environment. Complete physical-layer demodulation and state-machine transitions have been validated across continuous 50-burst evaluation batches, alongside physical acoustic over-the-air bench testing. The codebase is architected for rapid AUTOSAR C/C++ ECU migration and mapped directly to ISO 14229 UDS diagnostic parameters.

Commercial Opportunity & Turnkey Handover

This patent asset is offered for **outright assignment (sale) or commercial licensing**. The acquisition package provides complete freedom to adapt, rebrand, and integrate the technology across vehicle platforms with zero ongoing vendor dependencies:

- UK Patent Application No. GB2619997.6 (Full priority rights)
- 36-Page System Architecture & Threat Specification
- 350+ Page Engineering Implementation Dossier
- Reference Source Codebase (27 modular Csharp software modules)

Next Step: Private Demonstration Access

A 12-part video showcase of the working Software-in-the-Loop prototype is available on our password-protected evaluation portal.

Reply to this inquiry to receive the portal URL and secure access credentials.

Contact & MNDA Inquiries

John Murphy | Lead Inventor

Email: Mail@quackworkstudios.com

Full technical dossiers available under MNDA